- Operational status signals, such as an on-call flag or clearance level.

Which signals matter depends on the risk and the sensitivity of the decision. Viewing a document may depend solely on role and ownership. Deploying to production may also depend on device trust, time of day, and on-call status.

12.3.2 How signals are delivered

Signals can be transmitted through various methods, and the way they are delivered influences both effectiveness and confidence.

- **Token-based signals.** Some information is transmitted within tokens. Often in OAuth or OIDC ID tokens, but some services create custom JWTs to convey system-specific signals. These tokens carry signed claims about identity, device status, or authentication context. The benefit is speed: once the token is issued, the PDP doesn't need to query external systems. One risk is staleness if a claim changes before the token expires. Another is that tokens can be stolen and reused, so managing their expiration is critical to mitigate that risk.
- **Query-based signals**. Other signals are retrieved dynamically at decision time through APIs or databases. This guarantees freshness but introduces latency and dependencies on external systems.
- **Hybrid models**. Most organizations use a combination. Tokens provide a consistent identity and authentication context, while dynamic queries retrieve variable data such as risk scores or operational status.

The right mix depends on the system's requirements. Low-risk SaaS platforms usually depend solely on tokens. High-security infrastructure generally prefers hybrids.

12.3.3 ACME's delivery choices

At ACME, these tradeoffs are evident across its platforms:

- The Customer Collaboration platform mainly depends on token-based signals for customers. OAuth tokens from the customer authentication